2.2.4 Temperature dependence of resistivity

The resistivity of a material is dependent on temperature. It is experimentally found that for a wide range of temperatures, the resistivity of a conductor increases with increase in temperature according to the expression,

$$\rho_T = \rho_\circ\left[1+\alpha\left(T-T_\circ\right)\right] \qquad (2.27)$$

where ρ_T is the resistivity of a conductor at $T\,^\circ C$, ρ_o is the resistivity of the conductor at some reference temperature T_o (usually at 20°C) and α is the temperature coefficient of resistivity. **It is defined as the ratio of increase in resistivity per degree rise in temperature to its resistivity at T_o.**

From the equation (2.27), we can write

$$\rho_T - \rho_\circ = \alpha\rho_\circ\left(T-T_\circ\right)$$

$$\therefore \alpha = \frac{\rho_T-\rho_\circ}{\rho_\circ\left(T-T_\circ\right)} = \frac{\Delta\rho}{\rho_\circ\Delta T}$$

where $\Delta\rho = \rho_T - \rho_o$ is change in resistivity for a change in temperature $\Delta T = T - T_o$. Its unit is per °C.

α of conductors

For conductors α is positive. If the temperature of a conductor increases, the average kinetic energy of electrons in the conductor increases. This results in more frequent collisions and hence the resistivity increases. The graph of the equation (2.27) is shown in Figure 2.13.

Even though, the resistivity of conductors like metals varies linearly for wide range of temperatures, there also exists a non-linear region at very low temperatures. The resistivity approaches some finite value as the temperature approaches absolute zero as shown in Figure 2.13(b).

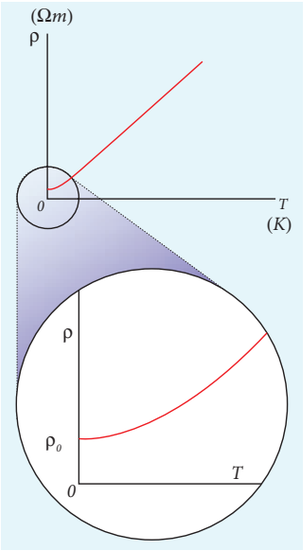


Figure 2.13 (a) Temperature dependence of resistivity for a conductor (b) Non linear region at low temperature

Using the equation $\rho = R\frac{A}{l}$ in equation (2.27), we get the expression for the resistance of a conductor at temperature T °C as

$$R_T = R_\circ\left[1+\alpha\left(T-T_\circ\right)\right] \qquad (2.28)$$

The temperature coefficient of resistivity can also be obtained from the equation (2.28),

$$R_T - R_\circ = \alpha R_\circ\left(T-T_\circ\right)$$

$$\therefore \alpha = \frac{R_T-R_\circ}{R_\circ\left(T-T_\circ\right)} = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T}$$

$$\alpha = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T} \qquad (2.29)$$

where $\Delta R = R_T - R_\circ$ is change in resistance during the change in temperature $\Delta T = T - T_\circ$

α of semiconductors

For semiconductors, the resistivity decreases with increase in temperature. As the temperature increases, more electrons will be liberated from their atoms (Refer unit 9 for conduction in semi conductors).

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